Uni-variate (1D) random variables (complete description)

Random variable	X
Outcome (value)	$X = x$
Probability density function (pdf)	$f_X(x) = \lim_{dx \to 0} \frac{P(X \in [x, x+dx])}{dx}$ $\int_{-\infty}^{\infty} f_X(x)dx = 1\,,\ f_X(x) \geq 0$ $P(X \in [a,b]) = \int_a^b f_X(x)dx$
Cumulative density function (cdf)	$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(x)dx$ $F_X(-\infty) = 0,\ \ F_X(\infty) = 1$ $P(X \in [a,b]) = F_X(b) - F_X(a)$
Domain and range	all random variables can be considered continuous with possible values $x \in R$, since discrete random variables can be modeled with impulse-functions in their pdf (and step-functions in their cdf) and since the pdf can equal zero in "impossible" x-regions

Population, i.e. theoretical, moments (summarizing, incomplete description)

Population mean (1'st order)	$\mu_X = E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)\,dx$ measures location (of probability mass)
Population variance (2'nd order)	$\sigma_X^2 = E[(X-\mu_X)^2] = \int_{-\infty}^{\infty} (x-\mu_X)^2 \cdot f_X(x)\,dx$ $\sigma_X^2 = E[X^2] - \mu_X^2$ measures dispersion (of probability mass)
Population standard deviation	$\sigma_X = \sqrt{\sigma_X^2}$
Population covariance	Cov(X,Y) = E [(X - mu_x)*(Y - mu_y)]

Descriptive statistics and sample moments

Sample (observed data values)	$X = \{x_1, x_2, \ldots x_n\}$
Sample size	n
Sample mean (average)	$\hat{\mu}_X = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
Sample variance	$\hat{\sigma}_X^2 = S_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$
Sample standard deviation	$\hat{\sigma}_X = \sqrt{\hat{\sigma}_X^2}$